

Replication Paper Quants II

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1 Introduction

This replication project, conducted as part of the Quantitative Methods II module, revisits the article “Personality Traits and Vote Switching: A Tale of Open-Minded Switchers and Agreeable Populism Resisters” by David Hamann and Sebastian Jäckle (Hamann and Jäckle 2025), published in *Politische Vierteljahresschrift*. The original study examines how the Big Five personality traits relate to (a) general vote switching and (b) switching specifically to the far-right populist AfD in Germany. Using the authors’ published replication data and code, I reproduce the paper’s main empirical outputs that rely on logistic regression (Figures 2-6) and the appendix regression tables (Tables A4 and A6). Because Figure 4 and the regression tables were originally implemented in STATA, I translated the corresponding STATA code into R to generate the replicated outputs. In addition to the replication, I extend the analysis in three ways: (1) I conduct basic diagnostic checks of the logistic regression models (multicollinearity and visual linearity-of-the-logit checks), (2) I estimate an interaction between openness and ideological self-placement for general vote switching (Model M1), and (3) I augment the AfD-switching model (Model M2) by adding a cultural ideology measure (GAL-TAN) to assess whether it improves explanatory power beyond the left-right scale. Finally, I use step-wise model selection to find the best model for M2.

2 Executive Summary

The replication largely confirms the substantive conclusions of Hamann and Jäckle (2025). The replicated figures and regression tables reproduce the same directional patterns and coefficient magnitudes as reported in the original paper. For general vote switching (Model M1), openness and conscientiousness are positively associated with switching, while extraversion is negatively associated; other traits show weaker or no robust effects. For susceptibility to populism (switching to the AfD; Model M2), ideological self-placement is by far the strongest predictor, and most Big Five coefficients are compar-

atively small and statistically weak in the baseline specification. Figure 4 could only be approximated visually due to the STATA-to-R translation, but the underlying predicted relationships are consistent with the published results.

The diagnostic extensions indicate that the core model assumptions are broadly reasonable: variance inflation factors are close to 1 in both models, suggesting no multicollinearity concerns, and the visual logit-linearity checks do not show strong non-linear patterns for the Big Five predictors. The interaction test (openness $\times$ left-right self-placement) is negative but not statistically significant, providing no clear evidence that openness operates differently across the ideological spectrum in predicting general vote switching. Finally, adding GAL-TAN to Model M2 improves fit substantially (AIC decreases from 1400.08 to 1330.33), and GAL-TAN is positive and highly significant, indicating that cultural ideological orientation explains additional variation in susceptibility to populism beyond the traditional left-right scale.

3 Original Paper

3.1 Goals of the Original Paper

The paper investigates whether stable personality differences help explain vote switching intentions in Germany. Building on the Big Five (OCEAN) framework, the authors focus on two outcomes: (a) general vote switching (any change in party preference across panel waves), and (b) switching specifically to the far-right populist AfD. They aim to test whether the Big Five Personality Factors predict switching directly, and whether any effects operate indirectly through voters' ideological self-placement on a left-right scale (used as a proxy for party identification). In doing so, the study extends prior work on personality and electoral volatility to a German context using multi-wave panel data and mediation analysis.

3.2 Hypotheses

The authors formulate two sets of hypotheses. For general vote switching, they expect that higher openness increases the likelihood of switching, while higher conscientiousness and extraversion reduce it; agreeableness is expected to increase switching, and no directional hypothesis is formulated for neuroticism.

For switching specifically to the far-right populist AfD, the authors expect that openness and agreeableness reduce the likelihood of switching, whereas conscientiousness and neuroticism increase it. They further propose that personality effects on switching to the AfD are substantially mediated by ideological left-right self-placement.

The analysis uses three waves of the Politikpanel Deutschland (PPD) online panel. Respondents who participated in all three waves, completed the surveys, reported vote intention, and were eligible to vote were included, yielding $N = 4351$ observations for the general vote-switching analysis. For the AfD-switching models, consistent AfD voters were excluded, resulting in $N = 4020$ observations. Because the panel is self-selected, the authors apply raking weights to match population benchmarks on age, gender, region, and vote intention.

3.3 Operationalisation

Personality is measured using the BFI-10 and aggregated into the five Big Five traits (John and Srivastava 1999): Openness to Experience describes the extent to which a person is imaginative, curious, and open to new ideas and experiences. Conscientiousness reflects how organized, disciplined, and goal-oriented an individual is. Extraversion captures the degree to which someone is sociable, energetic, and seeks stimulation in the company of others. Agreeableness refers to how compassionate, cooperative, and trusting a person tends to be toward others. Neuroticism indicates how prone an individual is to experiencing negative emotions such as anxiety, moodiness, and emotional instability.

Trait scores are averaged across items and waves and rescaled to range from 0 to 1.

Ideological self-placement on a left-right scale (also rescaled from 0 to 1) is used as a mediator because party identification is not available in the dataset. Control variables include age, gender, education, household income, and region.

3.4 Statistical Approach/Models

Hypotheses are tested (two-sided, $\alpha = 0.05$) using three complementary strategies: (1) weighted t-tests comparing personality means between groups, (2) weighted multiple logistic regressions including all Big Five traits, left-right self-placement, and controls, and (3) causal mediation analyses (separately for each trait), a linear model for the mediator and a logistic model for the outcome, with bootstrapped confidence intervals (1000 resamples) clustered at the individual level. The authors also report robustness checks (excluding outliers with highly volatile left-right placement, using controls from wave 3 instead of wave 1, and redefining the H2 control group to only non-AfD switchers).

3.5 Results

The results provide partial support for the authors' hypotheses. Regarding general vote switching, higher openness and - contrary to expectations - higher conscientiousness are significantly associated with a greater likelihood of switching. Extraversion and agreeableness show no consistent effects, while neuroticism exhibits only weak and unstable

associations. Mediation analyses indicate that the effects of openness and conscientiousness on general vote switching are primarily direct rather than operating through left-right self-placement.

For switching specifically to the AfD, openness and agreeableness reduce the likelihood of switching, whereas conscientiousness increases it; neuroticism shows mixed evidence. In contrast to the general switching models, most personality effects on switching to the AfD are substantially mediated by ideological left-right self-placement, suggesting that political orientation plays a central role in explaining susceptibility to far-right populism.

4 Replication

In a first step, I reproduced the main findings of the original paper using the provided data and code. A link to the replication repository can be found in the appendix. The full R script used for the replication is attached to this paper, while the specific code segments used to generate the replicated tables and figures are also included in the appendix. It should be noted that a large part of the original replication code is written in German.

The replication covers the main empirical results of the paper. Specifically, Figures 2 to 6 as well as Tables A4 and A6 were reproduced. Robustness checks reported in the original study were not replicated as part of this exercise.

4.1 Figures and Graphs

Table 1 from the original paper was not replicated, as it only contains descriptive statistics and is therefore less relevant for evaluating the core empirical results.

Figures 2, 3, 5, and 6 were successfully reproduced using the provided data and code without any major difficulties. The replicated figures below look identical to the original results reported in the paper.

Figure 4 required additional work. The original code used to produce this figure was written in STATA code. As Trinity does not provide a STATA license, the STATA code had to be translated into R. This translation did not work perfectly, and the resulting graph differs somewhat in appearance from the original figure. However, the underlying results could be reproduced in principle. The replicated figure shows the same substantive relationships reported in the original paper, even though the visual presentation is somewhat close to the original figure, but not identical. As the results are the same, I argue that this does represent a substantial replication.

Figure 1: Replication of Figure 3

Mean Values and Confidence Intervals of the Big Five by Party Preference

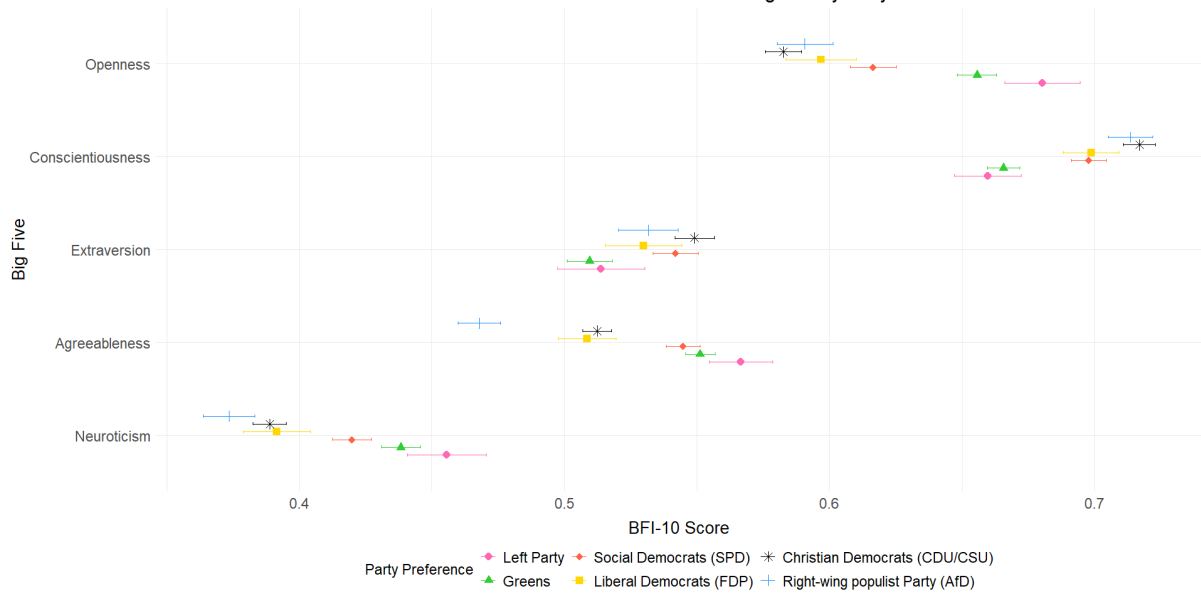


Figure 2: Replication of Figure 2

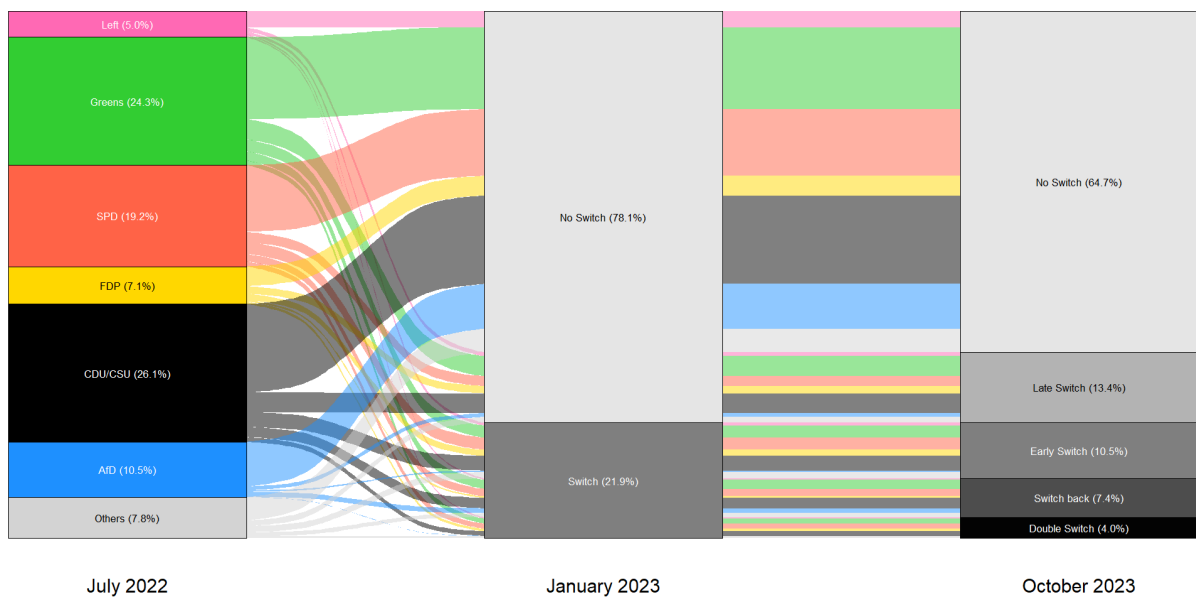


Figure 3: Replication of Figure 5

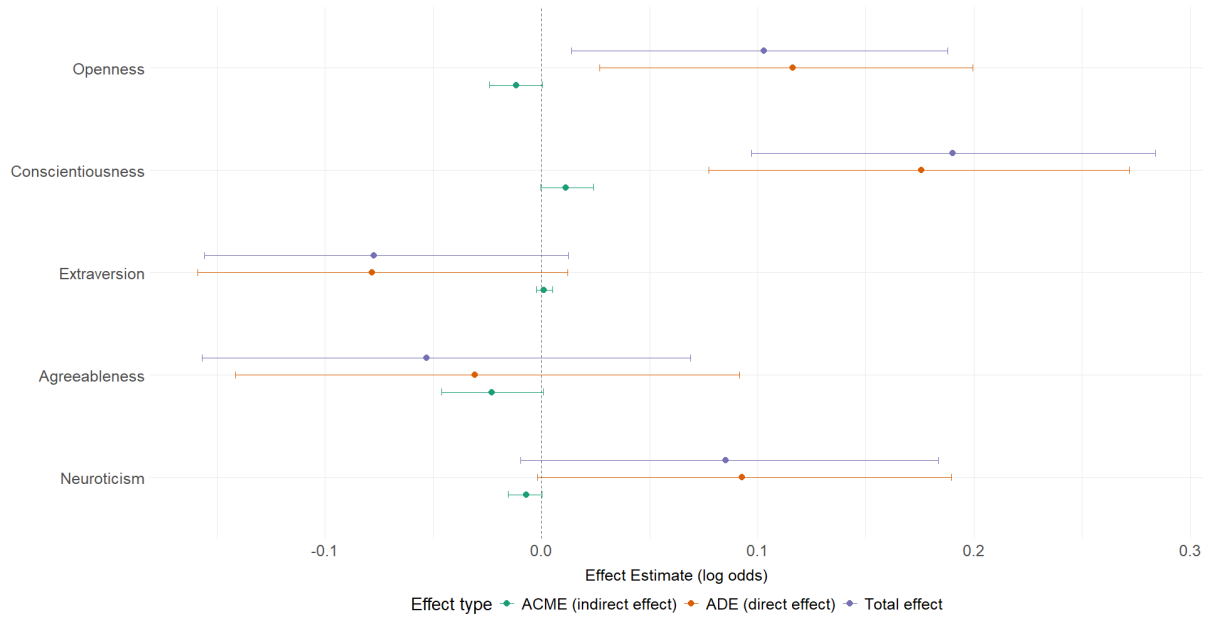


Figure 4: Replication of Figure 6

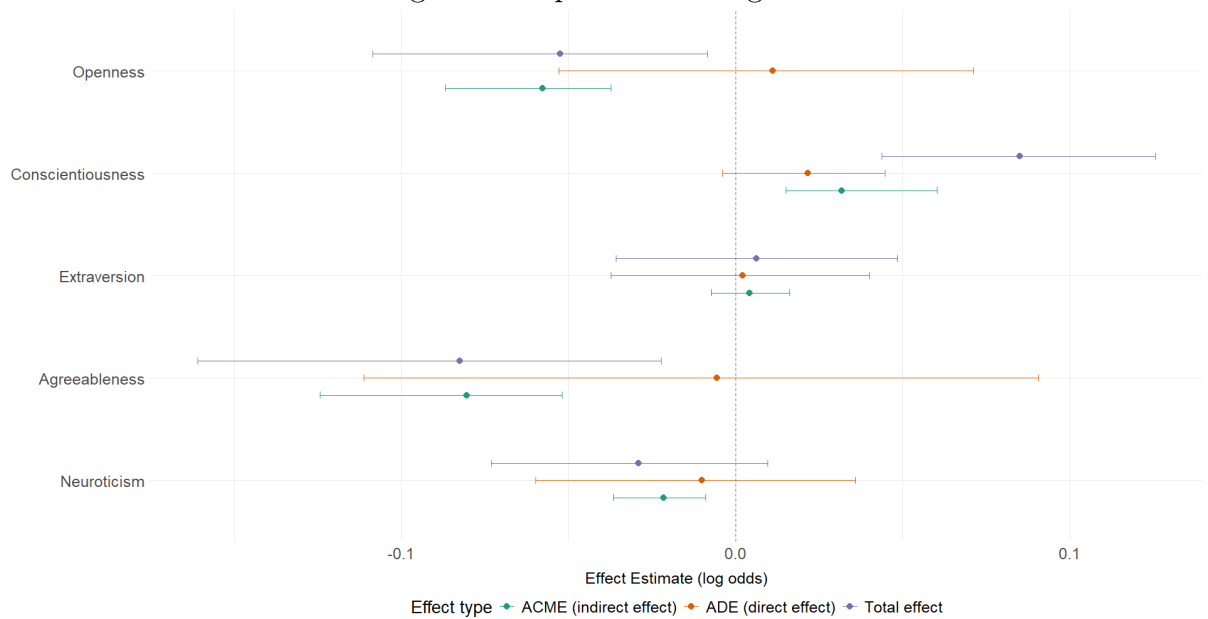


Figure 5: Original Figure 4

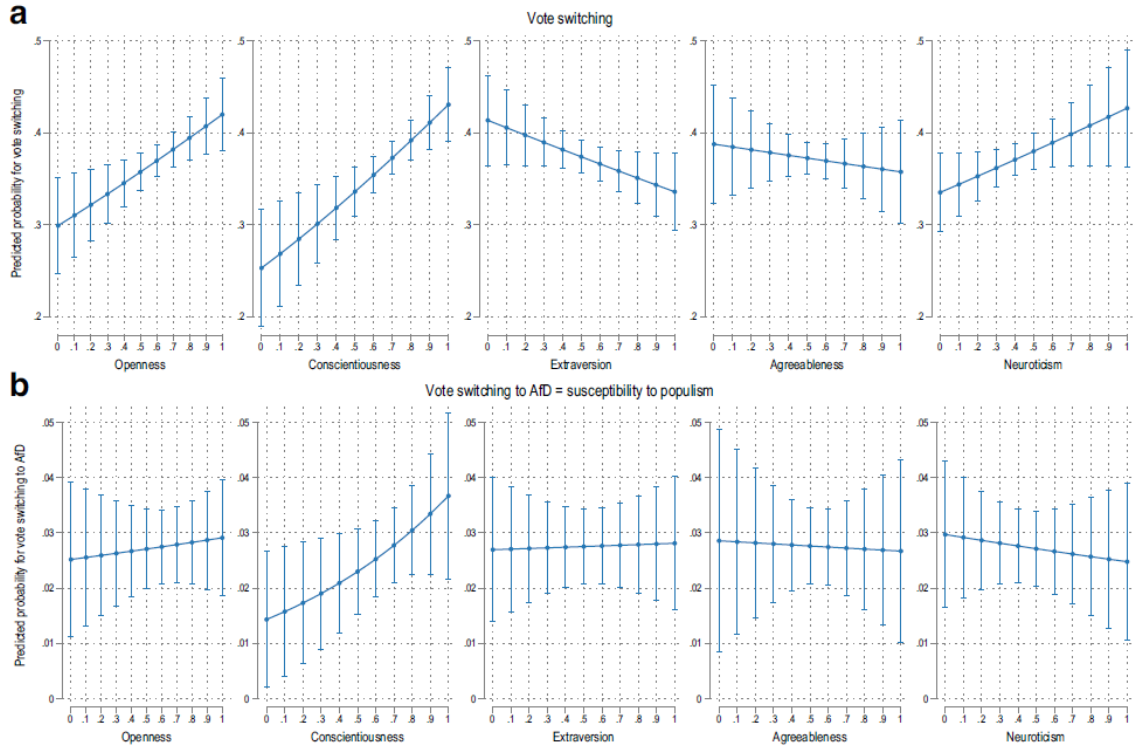
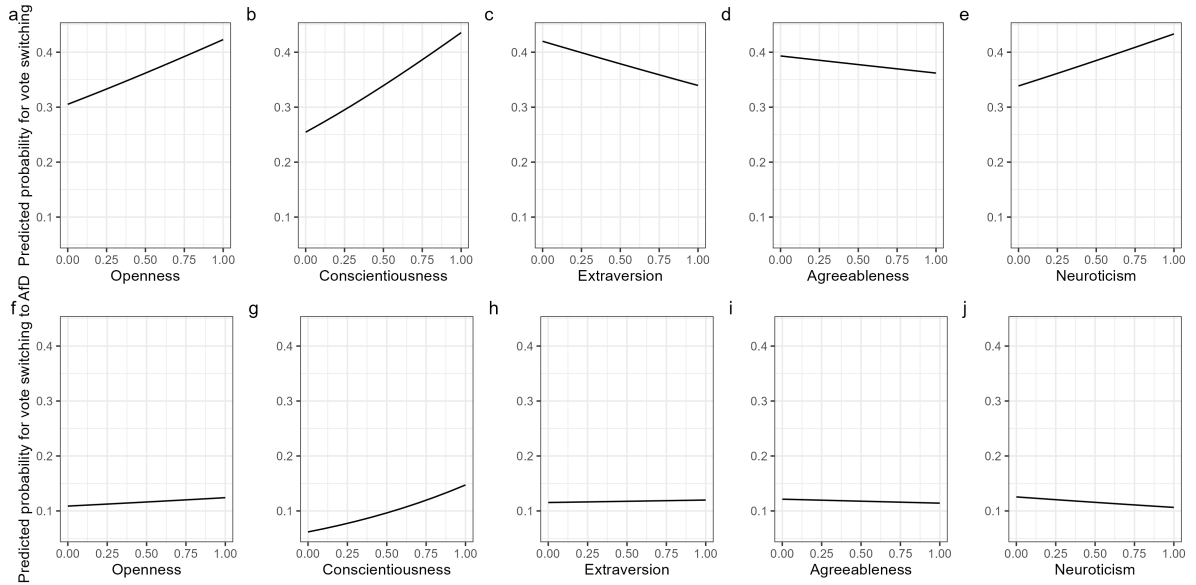


Figure 6: Replication of Figure 4



4.2 Tables

The most relevant tables for the replication exercise are Tables A4 and A6 from the appendix of the original paper. These tables present the logistic regression results for Model M1 (vote switching in general) and Model M2 (susceptibility to populism).

As with Figure 4, the original code used to generate these tables was written in STATA and therefore had to be translated into R. The replicated tables do not fully reproduce the exact formatting of the original tables, but the estimated coefficients, standard errors, and odds ratios correspond to those reported in the paper. In this sense, the replication can be considered successful with respect to the substantive results.

Some variable labels also had to be translated manually from German to English in order to match the terminology used in the original publication. The replicated tables are presented below, and the corresponding code used to generate them can be found in the appendix.

Table 1: Replication of Table A4. Logistic regression model M1: Vote switching in general

Variable	Coefficient (Standard Error)	Odds Ratio (95% CI)
Big Five traits		
Openness	0.5118 (0.1622) **	1.67 (1.21–2.29) **
Conscientiousness	0.8156 (0.2049) ***	2.26 (1.51–3.38) ***
Extraversion	-0.3422 (0.1571) *	0.71 (0.52–0.97) *
Agreeableness	-0.1325 (0.2053)	0.88 (0.59–1.31)
Neuroticism	0.4009 (0.1829) *	1.49 (1.04–2.14) *
Controls		
Left-right self-placement (0 = left, 1 = right)	0.3810 (0.1712) *	1.46 (1.05–2.05) *
Age	0.0023 (0.0021)	1.00 (1.00–1.01)
Gender (0 = male, 1 = female)	0.2107 (0.0682) **	1.23 (1.08–1.41) **
Education: ISCED 3	0.2747 (0.1721)	1.32 (0.94–1.84)
Education: ISCED 4	0.2913 (0.1744)	1.34 (0.95–1.88)
Education: ISCED 5–8	0.0291 (0.1662)	1.03 (0.74–1.43)
Income: I find it difficult to make ends meet	-0.1160 (0.2558)	0.89 (0.54–1.47)
Income: I find it fairly difficult to make ends meet	-0.0303 (0.2351)	0.97 (0.61–1.54)
Income: I find it fairly easy to make ends meet	-0.1903 (0.2333)	0.83 (0.52–1.31)
Income: I find it easy to make ends meet	-0.3058 (0.2418)	0.74 (0.46–1.18)
Region (0 = East Germany; 1 = West Germany)	0.0429 (0.0910)	1.04 (0.87–1.25)
Intercept	-1.7287 (0.3919) ***	0.18 (0.08–0.38) ***

Table 2: Replication of Table A6. Logistic regression model M2: Vote switching to AfD

Variable	Coefficient (Standard Error)	Odds Ratio (95% CI)
Big Five traits		
Openness	0.1488 (0.3660)	1.16 (0.57–2.38)
Conscientiousness	0.9639 (0.4649) *	2.62 (1.05–6.52) *
Extraversion	0.0435 (0.3541)	1.04 (0.52–2.09)
Agreeableness	-0.0696 (0.4491)	0.93 (0.39–2.25)
Neuroticism	-0.1863 (0.4178)	0.83 (0.37–1.88)
Controls		
Left-right self-placement (0 = left, 1 = right)	7.3578 (0.5286) ***	1568.37 (556.53–4419.87) ***
Age	0.0011 (0.0048)	1.00 (0.99–1.01)
Gender (0 = male, 1 = female)	-0.2199 (0.1557)	0.80 (0.59–1.09)
Education: ISCED 3	0.2853 (0.3695)	1.33 (0.64–2.74)
Education: ISCED 4	0.4878 (0.3759)	1.63 (0.78–3.40)
Education: ISCED 5–8	0.0590 (0.3638)	1.06 (0.52–2.16)
Income: I find it difficult to make ends meet	-0.7432 (0.4638)	0.48 (0.19–1.18)
Income: I find it fairly difficult to make ends meet	-1.2821 (0.4241) **	0.28 (0.12–0.64) **
Income: I find it fairly easy to make ends meet	-2.0401 (0.4285) ***	0.13 (0.06–0.30) ***
Income: I find it easy to make ends meet	-2.4671 (0.4630) ***	0.08 (0.03–0.21) ***
Region (0 = East Germany; 1 = West Germany)	-0.2549 (0.1969)	0.77 (0.53–1.14)
Intercept	-5.4391 (0.8291) ***	0.00 (0.00–0.02) ***

4.3 Replication Summary

Overall, the main empirical results of the original paper could be successfully reproduced. The replicated figures (Figures 2–6) and the key regression tables (Tables A4 and A6) show the same substantive patterns and coefficient estimates as reported in the original study. While minor differences in graphical presentation occurred, particularly for Figure

4 due to the translation of STATA code into R, the underlying relationships and statistical results remain consistent. Therefore, the replication confirms the central findings of the paper.

5 Extension

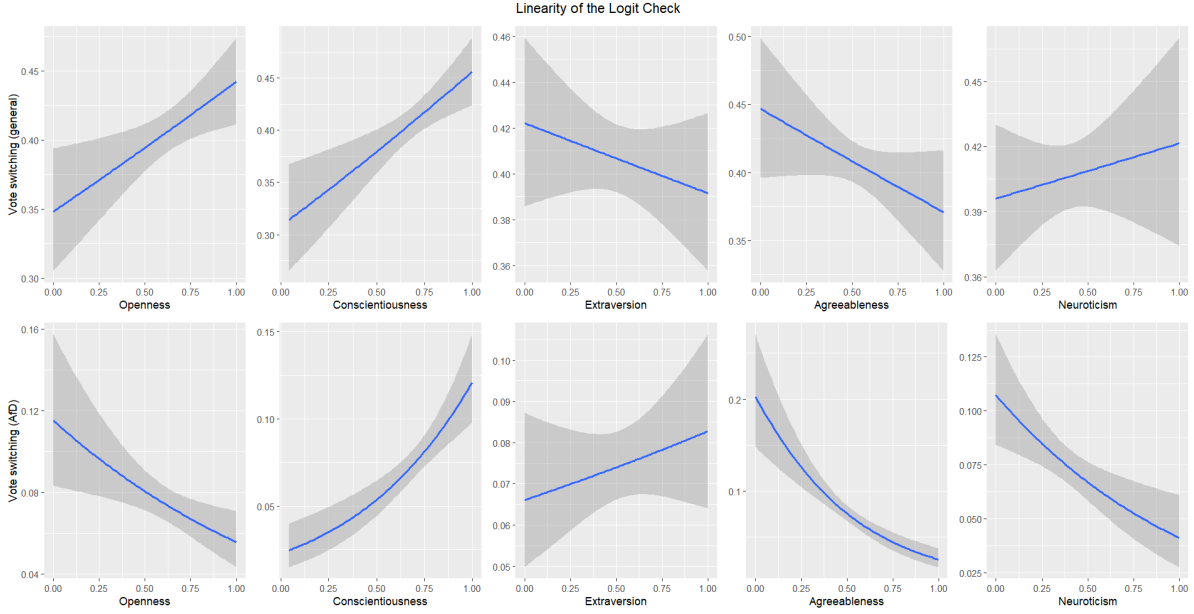
This chapter introduces three additions to the analyses in the main paper: a diagnostics checks, testing an interaction effect and testing the effects of including an additional control variable.

5.1 Diagnostic Checks

To assess whether the assumptions of the logistic regression models are met, two diagnostic checks were conducted. First, multicollinearity between the predictors was examined using variance inflation factors (VIF). For both models, all adjusted VIF values are close to 1 and well below commonly used thresholds of 5 or 10. In Model M1, the highest adjusted VIF is 1.10 (Neuroticism), while in Model M2 it is 1.09 (Extraversion). These values indicate that multicollinearity is not a concern and that the predictors are not strongly correlated with each other.

Second, the linearity of the logit assumption for the continuous predictors was assessed visually. Figure 7 below plots the estimated logistic relationship between each of the Big Five personality traits and the probability of vote switching, both generally (Model M1) and specifically to the AfD (Model M2). The fitted curves for most predictors show broadly monotonic relationships across the observed range of the personality scales. Although some curvature is visible in a few cases, particularly for agreeableness and conscientiousness in Model M2, the general patterns remain approximately linear over most of the observed range. Overall, the plots do not indicate substantial violations of the linearity of the logit assumption, suggesting that the logistic regression specification is appropriate.

Figure 7: Linearity Check



Overall, the diagnostic checks indicate that the logistic regression models are appropriately specified and that the estimated coefficients can be interpreted reliably. All used R code for this can be found in the appendix.

5.2 Interaction Effect

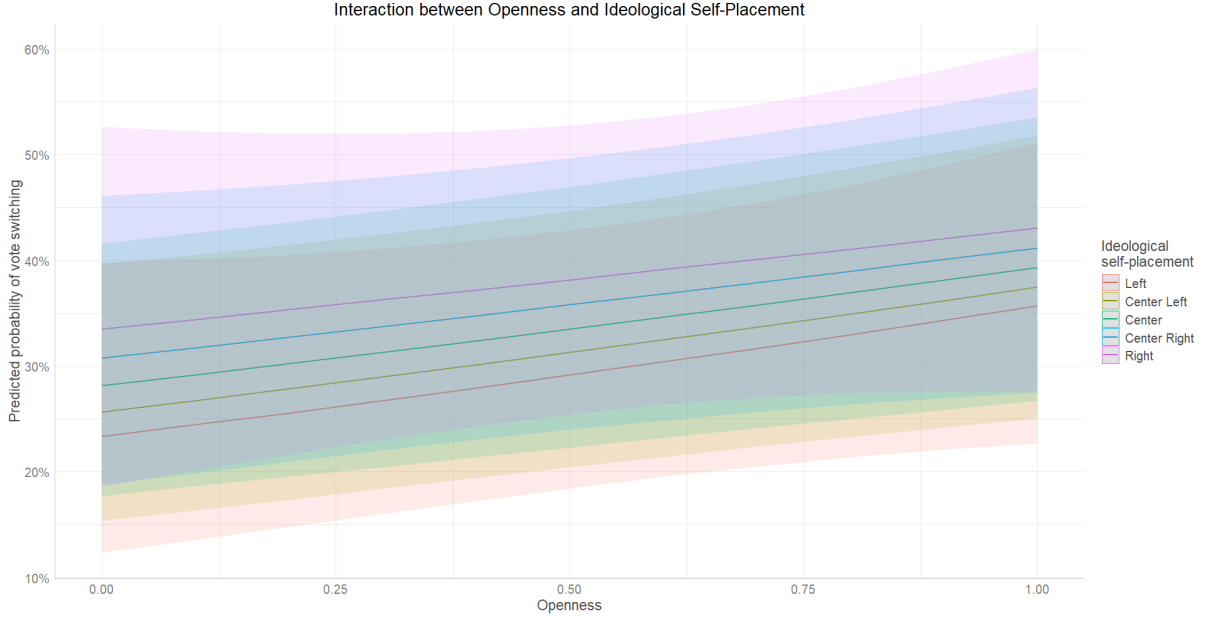
As a first extension to the original analysis, an interaction between openness and ideological self-placement was estimated to examine whether the effect of openness on vote switching depends on voters' ideological orientation. This extension was applied only to the model of vote switching in general (Model M1), as the theoretical rationale is less applicable to vote switching specifically to the AfD.

A potential interaction between openness and ideological self-placement may arise because the political implications of openness depend on individuals' broader ideological orientation (Van Hiel, Kossowska, and Mervielde 2000). Openness is commonly associated with receptiveness to new ideas, tolerance, and a preference for social and political change. Among individuals who identify with more left-leaning ideological positions, higher levels of openness may reinforce support for alternative or progressive political options and thus increase the likelihood of switching parties. In contrast, among individuals with more right-leaning ideological orientations, ideological commitments may constrain how personality traits translate into political behaviour, potentially weakening the relationship between openness and vote switching.

To test this possibility, an interaction term between openness and ideological self-placement was included in Model M1. The estimated interaction effect is negative but not statistically significant ($\beta = -0.19$, $p = 0.80$), indicating no clear evidence that the effect of

openness differs systematically across ideological positions. Although the interaction plot suggests that the association between openness and vote switching may be somewhat weaker among individuals who place themselves further to the right, these differences are not statistically distinguishable from zero. Figure 8 illustrates the predicted probabilities derived from the interaction model.

Figure 8: Interaction effect between openness and ideological self-placement



5.3 Additional Variable

As an additional extension, the cultural ideology dimension (GAL–TAN) was included in the model predicting susceptibility to populism. While the original specification controls for respondents’ left–right self-placement, a dimension that primarily captures economic ideology. However, research on populist and especially right-wing populist attitudes suggests that cultural conflict, often summarized by the GAL–TAN (Green–Alternative–Libertarian vs. Traditional–Authoritarian–Nationalist) dimension, also plays an important role in explaining support for such positions (Kurella and Rapp 2026).

Including the GAL–TAN variable therefore allows us to test whether susceptibility to populism (in the sense of switching their vote to the AfD) is associated not only with economic ideology but also with cultural ideological orientations. This provides a more comprehensive specification of the model and allows us to examine whether the effects observed in the original model remain robust once cultural ideology is taken into account. The results of the extended model are largely consistent with the baseline specification. The GAL–TAN dimension is positive and highly significant ($\beta = 1.69$, $p < 0.01$), indicating that individuals with more traditional, authoritarian, and nationalist orientations are more likely to switch their vote to the AfD. At the same time, the main findings

of the original model remain stable: conscientiousness continues to show a positive and significant association, while the other personality traits remain insignificant.

Table 3: Logistic Regression Models for Susceptibility to Populism

	<i>Dependent variable: Susceptibility to populism</i>		
	M2 (1)	M2 + GAL-TAN (2)	Selected Model (3)
Openness	0.315 (0.373)	0.465 (0.375)	
Conscientiousness	0.899* (0.471)	0.973** (0.472)	1.061** (0.442)
Extraversion	-0.007 (0.364)	-0.069 (0.367)	
Agreeableness	-0.151 (0.456)	-0.127 (0.456)	
Neuroticism	-0.111 (0.426)	-0.066 (0.426)	
Left-right self-placement	7.250*** (0.536)	5.978*** (0.621)	5.924*** (0.610)
GAL-TAN		1.686*** (0.430)	1.674*** (0.427)
Age	0.001 (0.005)	-0.00001 (0.005)	
Gender (female)	-0.259 (0.160)	-0.283* (0.160)	-0.284* (0.156)
Education: ISCED 3	0.300 (0.370)	0.367 (0.371)	0.363 (0.365)
Education: ISCED 4	0.455 (0.377)	0.544 (0.378)	0.551 (0.367)
Education: ISCED 5-8	-0.024 (0.365)	0.069 (0.366)	0.108 (0.358)
Income: difficult to make ends meet	-0.675 (0.479)	-0.773 (0.487)	-0.794 (0.484)
Income: fairly difficult to make ends meet	-1.181*** (0.438)	-1.229*** (0.446)	-1.276*** (0.442)
Income: fairly easy to make ends meet	-1.904*** (0.442)	-1.944*** (0.450)	-1.998*** (0.447)
Income: easy to make ends meet	-2.301*** (0.476)	-2.326*** (0.484)	-2.376*** (0.479)
Region (West Germany)	-0.261 (0.200)	-0.199 (0.201)	
Constant	-5.431*** (0.844)	-5.915*** (0.860)	-5.933*** (0.661)
Observations	3,816	3,816	3,816
Log Likelihood	-656.041	-647.165	-648.355
Akaike Inf. Crit.	1,346.082	1,330.330	1,320.710

Note: * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$

A likelihood ratio test comparing the baseline model and the extended model including the GAL–TAN variable shows that the latter significantly improves model fit ($\chi^2(1) = 15.62$, $p < 0.001$). This suggests that cultural ideological orientation provides additional explanatory power for susceptibility to populism beyond the traditional left–right scale.

After this, a stepwise model selection procedure was conducted to identify the specification that provides the best fit to the data. Table 3 below thus compares three models: the original Model 2, an extended model that includes the GAL–TAN measure from the original dataset, and the final model selected through the stepwise procedure.

The results shown in Table 3 indicate that across all specifications, conscientiousness remains positively and significantly associated with susceptibility to populism (vote switching to AfD), while the other personality traits remain insignificant. The stepwise selected model retains the key predictors conscientiousness, left–right self-placement, and the GAL–TAN dimension, indicating that both economic and cultural ideological orientations play an important role in explaining susceptibility to populism.

References

- Hamann, David and Sebastian Jäckle (Oct. 29, 2025). “Personality Traits and Vote Switching: A Tale of Open-Minded Switchers and Agreeable Populism Resisters”. In: *Politische Vierteljahresschrift*. ISSN: 0032-3470, 1862-2860. DOI: 10.1007/s11615-025-00637-3. URL: <https://link.springer.com/10.1007/s11615-025-00637-3> (visited on 03/30/2026).
- John, Oliver P. and Sanjay Srivastava (1999). “The Big Five Trait taxonomy: History, measurement, and theoretical perspectives.” In: *Handbook of personality: Theory and research*, 2nd ed. New York, NY, US: Guilford Press, pp. 102–138. ISBN: 1-57230-483-9 (Hardcover).
- Kurella, Anna-Sophie and Milena Rapp (Apr. 16, 2026). “Unfolding GAL-TAN: the multi-dimensional nature of public opinion in Western Europe”. In: *West European Politics* 49.3, pp. 841–866. ISSN: 0140-2382, 1743-9655. DOI: 10.1080/01402382.2025.2466117. URL: <https://www.tandfonline.com/doi/full/10.1080/01402382.2025.2466117> (visited on 03/30/2026).
- Van Hiel, Alain, Malgorzata Kossowska, and Ivan Mervielde (Apr. 2000). “The relationship between Openness to Experience and political ideology”. In: *Personality and Individual Differences* 28.4, pp. 741–751. ISSN: 01918869. DOI: 10.1016/S0191-8869(99)00135-X. URL: <https://linkinghub.elsevier.com/retrieve/pii/S019188699900135X> (visited on 03/30/2026).

6 Appendix

Paper available (open access) on Springer Nature here: <https://link.springer.com/article/10.1007/s11615-025-00637-3>

Replication code and data available on Harvard Dataverse here: <https://dataverse.harvard.edu/dataset.xhtml?persistentId=doi:10.7910/DVN/OLWMMF>

The appendix of the original paper can be found here: https://static-content.springer.com/esm/art%3A10.1007%2Fs11615-025-00637-3/MediaObjects/11615_2025_637_MOESM1_ESM.pdf

Documentation for attached files: Figures 2, 3, 5, and 6 were replicated using the original replication script "Personality Traits and Vote Swing_v2.R". Figure 4 was originally implemented in STATA and provided in the file "Figure 4.do" of the original replication data. As STATA was not available, I translated the code into R and implemented it through the script "Figure 4.R".

Tables A4 and A6 were replicated using my own R code, partly based on translated snippets from the original STATA file "Figure 4.do". The table replication, together with the additional model extensions presented in this paper, can be found in the script "Table replication and Extension.R".

Listing 1: Replication code for Figure 3

```

1 # Erstellung des Plots mit Konfidenzintervallen
2 p2 <- ggplot(big_five_long_ci, aes(x = mean, y = trait, color =
3   Parteipräferenz, shape = Parteipräferenz)) +
4   geom_point(position = position_dodge(width = 0.5), size = 3) +
5   geom_errorbarh(aes(xmin = mean - 1.96 * se, xmax = mean + 1.96 * se),
6     position = position_dodge(width = 0.5), height = 0.2)
7   +
8   labs(title = "Mean Values and Confidence Intervals of the Big Five by
9     Party Preference",
10     x = "BFI-10 Score",
11     y = "Big Five",
12     color = "Party Preference",
13     shape = "Party Preference") +
14   scale_color_manual(values = c("Linke" = "#FF69B4",
15     "Grüne" = "#32CD32",
16     "SPD" = "#FF6347",
17     "FDP" = "#FFD700",
18     "Union" = "black",
19     "AfD" = "#1E90FF",
20     "Andere" = "lightgrey"),
21     labels = c("Linke" = "Left Party",
22     "Grüne" = "Greens",
23     "SPD" = "Social Democrats (SPD)",
24     "FDP" = "Liberal Democrats (FDP)",
25     "Union" = "Christian Democrats (CDU/CSU
26   )",
27     "AfD" = "Right-wing populist Party (AfD
28   )",
29     "Andere" = "Other")) +
30   scale_shape_manual(values = c("Linke" = 16, # Kreis
31     "Grüne" = 17, # Dreieck
32     "SPD" = 18, # Raute
33     "FDP" = 15, # Quadrat
34     "Union" = 8, # Stern
35     "AfD" = 3, # Kreuz
36     "Andere" = 3), # Plus
37     labels = c("Linke" = "Left Party",
38     "Grüne" = "Greens",
39     "SPD" = "Social Democrats (SPD)",
40     "FDP" = "Liberal Democrats (FDP)",
41     "Union" = "Christian Democrats (CDU/CSU
42   )",
43     "AfD" = "Right-wing populist Party (AfD
44   )",
45     "Andere" = "Other")) +
46   theme_minimal(base_family = "Times") +

```

```

40 theme(
41   plot.title = element_text(hjust = 0.5, family = "Times", size = 18)
42   ,
43   axis.title = element_text(family = "Times", size = 16),
44   axis.title.x = element_text(vjust = -1),
45   axis.text = element_text(family = "Times", size = 14),
46   legend.text = element_text(family = "Times", size = 14),
47   legend.title = element_text(family = "Times", size = 14),
48   legend.position = "bottom")
49 # Plot ausgeben
50 print(p2)

```

Listing 2: Replication code for Figure 2

```

1 # Sankey-Diagramm erstellen
2 ggplot(sankey_data,
3       aes(axis1 = welle_7, axis2 = w7_w8, axis3 = w7_w8_w9, y =
4         Gewicht)) +
5
6 # Flüsse in Parteifarben
7
8 # Strata: Achse 1 bunt, Achsen 2/3 Grautöne
9 geom_stratum(
10   aes(fill = ifelse(..x.. == 1, as.character(..stratum..), as.
11     character(..stratum..))),
12   width = 0.5, color = "black"
13 ) +
14
15 # Strata-Beschriftungen mit Prozenten
16 geom_text(
17   stat = "stratum",
18   aes(
19     label = ifelse(..x.. == 1,
20       paste0(..stratum.., " (", sprintf("%.1f", 100 * ..
21         prop..), "%)"),
22       paste0(..stratum.., " (", sprintf("%.1f", 100 * ..
23         prop..), "%)"),
24     color = ifelse(..stratum.. %in% c("FDP", "Others", "No Switch", "
25       Late Switch"), "black", "white")
26   ),
27   size = 3.5, family = "sans"
28 ) +
29 scale_color_identity() +
30

```

```

28 scale_x_discrete(
29   limits = c("July 2022", "January 2023", "October 2023"),
30   expand = c(.05, .05)
31 ) +
32
33 scale_fill_manual(
34   values = c(party_colors, switch_colors, "Andere" = "grey80"),
35   na.value = "grey50"
36 ) +
37
38 theme_minimal(base_family = "sans") +
39 theme(
40   panel.grid = element_blank(),
41   axis.title = element_blank(),
42   axis.text.y = element_blank(),
43   axis.ticks = element_blank(),
44   axis.text.x = element_text(size = 18, color = "black", margin =
margin(t = 10)),
45   legend.position = "none"
46 )

```

Listing 3: Replication code for Figure 5

```

1 # Plot
2 ggplot(results_df_plot, aes(x = estimate, y = variable, color = effect))
3   +
4   geom_point(position = position_dodge(width = 0.5), size = 2.5) +
5   geom_errorbar(aes(xmin = ci.low, xmax = ci.high),
6                 position = position_dodge(width = 0.5), width = 0.2) +
7   geom_vline(xintercept = 0, linetype = "dashed", color = "gray40") +
8   scale_color_manual(
9     values = c("ACME" = "#1b9e77", "ADE" = "#d95f02", "Total Effect" =
"#7570b3"),
10    labels = c("ACME (indirect effect)", "ADE (direct effect)", "Total
effect")
11 ) +
12 labs(
13   x = "Effect Estimate (log odds)",
14   y = NULL,
15   color = "Effect type"
16 ) +
17 theme_minimal(base_family = "Times") +
18 theme(
19   legend.position = "bottom",
20   legend.title = element_text(size = 17),
21   legend.text = element_text(size = 15),
22   axis.title.x = element_text(vjust = -1, size = 15),

```

```

22   axis.title.y = element_text(hjust = 0.5, size = 15),
23   axis.text = element_text(size = 15)
24 )

```

Listing 4: Replication code for Figure 6

```

1  # Plot
2  ggplot(results_df_plot2, aes(x = estimate, y = variable, color = effect
   )) +
3    geom_point(position = position_dodge(width = 0.5), size = 2.5) +
4    geom_errorbar(aes(xmin = ci.low, xmax = ci.high),
5                  position = position_dodge(width = 0.5), width = 0.2) +
6    geom_vline(xintercept = 0, linetype = "dashed", color = "gray40") +
7    scale_color_manual(
8      values = c("ACME" = "#1b9e77", "ADE" = "#d95f02", "Total Effect" =
9        "#7570b3"),
10     labels = c("ACME (indirect effect)", "ADE (direct effect)", "Total
11       effect")
12   ) +
13   labs(
14     x = "Effect Estimate (log odds)",
15     y = NULL,
16     color = "Effect type"
17   ) +
18   theme_minimal(base_family = "Times") +
19   theme(
20     legend.position = "bottom",
21     legend.title = element_text(size = 17),
22     legend.text = element_text(size = 15),
23     axis.title.x = element_text(vjust = -1, size = 15),
24     axis.title.y = element_text(hjust = 0.5, size = 15),
25     axis.text = element_text(size = 15)
26   )

```

Listing 5: Replication code for Tables 1 and 2 (Tables A4 and A6 in the original

```

1  rm(list = ls())
2
3  setwd("C:/Users/onnos/Desktop/TRINITY/PhD/Quantitative Methods II/
   Replication Paper/Replication")
4
5  # packages
6  library(haven)
7  library(dplyr)
8  library(ggplot2)
9  library(patchwork)
10
11

```

```

12 # =====
13 # PART 1: Vote switching (ww); Stata section 1
14 # =====
15
16 df1 <- read_dta("data/data_wahlwechsel.dta") %>%
17   mutate(
18     ww = case_when(
19       Wahlwechsel == 1 ~ 0,
20       Wahlwechsel == 2 ~ 1,
21       TRUE ~ NA_real_
22     ),
23     geschlecht = factor(geschlecht),
24     bildung = factor(bildung),
25     haushaltseinkommen = factor(haushaltseinkommen),
26     region = factor(region)
27   ) %>%
28   # important: match Stata estimation sample (complete cases)
29   filter(complete.cases(
30     ww, OpennessMittelwert, ConscientiousnessMittelwert,
31     ExtraversionMittelwert,
32     AgreeablenessMittelwert, NeuroticismMittelwert, LiRe, alter,
33     geschlecht, bildung, haushaltseinkommen, region, ipf_gewicht
34   ))
35
36 m1 <- glm(
37   ww ~ OpennessMittelwert + ConscientiousnessMittelwert +
38     ExtraversionMittelwert +
39     AgreeablenessMittelwert + NeuroticismMittelwert + LiRe + alter +
40     geschlecht + bildung + haushaltseinkommen + region,
41   data = df1,
42   family = binomial(link = "logit"),
43   weights = ipf_gewicht
44 )
45 # =====
46 # PART 2: Populism (pop); Stata section 2
47 # =====
48
49 df2 <- read_dta("data/data_populismus.dta") %>%
50   mutate(
51     pop = case_when(
52       Populismusanfällige == 1 ~ 0,
53       Populismusanfällige == 2 ~ 1,
54       TRUE ~ NA_real_
55     ),
56     geschlecht = factor(geschlecht),

```

```

57   bildung = factor(bildung),
58   haushaltseinkommen = factor(haushaltseinkommen),
59   region = factor(region)
60 ) %>%
61 filter(complete.cases(
62   pop, OpennessMittelwert, ConscientiousnessMittelwert,
63   ExtraversionMittelwert,
64   AgreeablenessMittelwert, NeuroticismMittelwert, LiRe, alter,
65   geschlecht, bildung, haushaltseinkommen, region, ipf_gewicht
66 ))
67 m2 <- glm(
68   pop ~ OpennessMittelwert + ConscientiousnessMittelwert +
69   ExtraversionMittelwert +
70   AgreeablenessMittelwert + NeuroticismMittelwert + LiRe + alter +
71   geschlecht + bildung + haushaltseinkommen + region,
72   data = df2,
73   family = binomial(link = "logit"),
74   weights = ipf_gewicht
75 )
76 # =====
77 # PART 3:The table
78 # =====
79
80 labels_en <- c(
81   "(Intercept)" = "Intercept",
82
83   # Big Five
84   "OpennessMittelwert" = "Openness",
85   "ConscientiousnessMittelwert" = "Conscientiousness",
86   "ExtraversionMittelwert" = "Extraversion",
87   "AgreeablenessMittelwert" = "Agreeableness",
88   "NeuroticismMittelwert" = "Neuroticism",
89
90   # Controls
91   "LiRe" = "Left-right self-placement (0 = left, 1 = right)",
92   "alter" = "Age",
93   "geschlecht2" = "Gender (0 = male, 1 = female)",
94   "region2" = "Region (0 = East Germany; 1 = West Germany)",
95
96   # Education (ref: ISCED 0-2)
97   "bildung2" = "Education: ISCED 3",
98   "bildung3" = "Education: ISCED 4",
99   "bildung4" = "Education: ISCED 5-8",
100
101   # Income (ref: I find it very difficult to make ends meet)

```

```

102 "haushaltseinkommen2" = "Income: I find it difficult to make ends
    meet",
103 "haushaltseinkommen3" = "Income: I find it fairly difficult to make
    ends meet",
104 "haushaltseinkommen4" = "Income: I find it fairly easy to make ends
    meet",
105 "haushaltseinkommen5" = "Income: I find it easy to make ends meet")
106
107 sig <- function(p) ifelse(p<.001,"***",
108                             ifelse(p<.01,"**",
109                                     ifelse(p<.05,"*", "")))
110
111 make_a_table <- function(m, labels_en, caption){
112
113   df <- broom::tidy(m) %>%
114     dplyr::mutate(
115       Variable = ifelse(term %in% names(labels_en), labels_en[term],
116                         term),
117       stars = sig(p.value),
118       'Coefficient (Standard Error)' = sprintf("%.4f (%.4f) %s",
119         estimate, std.error, stars),
120       OR = exp(estimate),
121       lo = exp(estimate - 1.96*std.error),
122       hi = exp(estimate + 1.96*std.error),
123       'Odds Ratio (95\\% CI)' = sprintf("%.2f (%.2f--%.2f) %s", OR, lo,
124         hi, stars)
125     ) %>%
126     dplyr::select(Variable, 'Coefficient (Standard Error)', 'Odds Ratio
127       (95\\% CI)')
128
129 # section headers like in the appendix
130 big5 <- c("Openness","Conscientiousness","Extraversion","
131   Agreeableness","Neuroticism")
132
133 df2 <- df %>%
134   dplyr::mutate(section = dplyr::case_when(
135     Variable %in% big5 ~ "Big Five traits",
136     Variable == "Intercept" ~ "Intercept",
137     TRUE ~ "Controls"
138   )) %>%
139   dplyr::group_by(section) %>%
140   dplyr::group_modify(~{
141     if (.y$section %in% c("Big Five traits","Controls")) {
142       dplyr::bind_rows(
143         dplyr::tibble(
144           Variable = .y$section,
145           'Coefficient (Standard Error)' = "",

```

```

141         'Odds Ratio (95\\% CI)' = ""
142     ),
143     .x
144 )
145 } else {
146     .x
147 }
148 }) %>%
149 dplyr::ungroup() %>%
150 dplyr::select(-section)
151
152 print(
153     xtable(df2, caption = caption, align = c("l", "p{7cm}", "l", "l")),
154     include.rownames = FALSE,
155     sanitize.text.function = identity,
156     comment = FALSE
157 )
158 }
159
160 # --- create and print tables ---
161 make_a_table(m1, labels_en, "Table A4. Logistic regression model M1 --
    Vote switching in general")
162 make_a_table(m2, labels_en, "Table A6. Logistic regression model M2 --
    Vote switching to AfD (susceptibility to populism)")

```

Listing 6: Replication code for the Diagnostic Checks

```

1 # =====
2 # Extension 1: Model Assumptions
3 # =====
4 library(stargazer)
5 vif(m1)
6 vif(m2)
7
8 p1 <- ggplot(df1, aes(OpennessMittelwert, ww)) +
9     geom_smooth(method = "glm", method.args = list(family = "binomial"))
10     +
11     labs(x = "Openness", y = "Vote switching (general)")
12
13 p2 <- ggplot(df1, aes(ConscientiousnessMittelwert, ww)) +
14     geom_smooth(method = "glm", method.args = list(family = "binomial"))
15     +
16     labs(x = "Conscientiousness", y = NULL)
17
18 p3 <- ggplot(df1, aes(ExtraversionMittelwert, ww)) +
19     geom_smooth(method = "glm", method.args = list(family = "binomial"))
20     +

```

```

18   labs(x = "Extraversion", y = NULL)
19
20 p4 <- ggplot(df1, aes(AgreeablenessMittelwert, ww)) +
21   geom_smooth(method = "glm", method.args = list(family = "binomial"))
22   +
23   labs(x = "Agreeableness", y = NULL)
24
25 p5 <- ggplot(df1, aes(NeuroticismMittelwert, ww)) +
26   geom_smooth(method = "glm", method.args = list(family = "binomial"))
27   +
28   labs(x = "Neuroticism", y = NULL)
29
30 p6 <- ggplot(df2, aes(OpennessMittelwert, pop)) +
31   geom_smooth(method = "glm", method.args = list(family = "binomial"))
32   +
33   labs(x = "Openness", y = "Vote switching (AfD)")
34
35 p7 <- ggplot(df2, aes(ConscientiousnessMittelwert, pop)) +
36   geom_smooth(method = "glm", method.args = list(family = "binomial"))
37   +
38   labs(x = "Conscientiousness", y = NULL)
39
40 p8 <- ggplot(df2, aes(ExtraversionMittelwert, pop)) +
41   geom_smooth(method = "glm", method.args = list(family = "binomial"))
42   +
43   labs(x = "Extraversion", y = NULL)
44
45 p9 <- ggplot(df2, aes(AgreeablenessMittelwert, pop)) +
46   geom_smooth(method = "glm", method.args = list(family = "binomial"))
47   +
48   labs(x = "Agreeableness", y = NULL)
49
50 p10 <- ggplot(df2, aes(NeuroticismMittelwert, pop)) +
51   geom_smooth(method = "glm", method.args = list(family = "binomial"))
52   +
53   labs(x = "Neuroticism", y = NULL)
54
55 fig <- (p1 | p2 | p3 | p4 | p5) /
56   (p6 | p7 | p8 | p9 | p10) +
57   plot_annotation(
58     title = "Linearity of the Logit Check",
59     theme = theme(plot.title = element_text(hjust = 0.5)))
60 print(fig)
61 ggsave("figures/linearity_check.pdf", fig, width = 12, height = 3)

```

Listing 7: Replication Code for the Interaction Effect

```

1 # =====
2 # Extension 2: Interaction Effects
3 # =====
4
5 m1_int <- glm(
6   ww ~ OpennessMittelwert * LiRe +
7     ConscientiousnessMittelwert + ExtraversionMittelwert +
8     AgreeablenessMittelwert + NeuroticismMittelwert +
9     alter + geschlecht + bildung + haushaltseinkommen + region,
10  data = df1,
11  family = binomial(link = "logit"),
12  weights = ipf_gewicht
13 )
14 summary(m1_int)
15 pred <- ggpredict(m1_int,
16                   terms = c("OpennessMittelwert",
17                             "LiRe [0.0,0.25,0.5,0.75,1]"))
18
19 plot(pred) +
20   labs(
21     title = "Interaction between Openness and Ideological Self-
22             Placement",
23     x = "Openness",
24     y = "Predicted probability of vote switching",
25     color = "Ideological
26 self-placement"
27   ) +
28   scale_color_discrete(labels = c("Left", "Center Left", "Center", "
29                                   Center Right", "Right")) +
30   theme(
31     plot.title = element_text(hjust = 0.5, size = 16),
32     axis.title = element_text(size = 14),
33     axis.text = element_text(size = 12),
34     legend.title = element_text(size = 14),
35     legend.text = element_text(size = 12))

```

Listing 8: Replication code for the Added Variable

```

1 # =====
2 # Extension 3: Added Variable
3 # =====
4
5 df2 <- read_dta("data/data_populismus.dta") %>%
6   mutate(
7     pop = case_when(
8       Populismusanfällige == 1 ~ 0,
9       Populismusanfällige == 2 ~ 1,

```

```

10     TRUE ~ NA_real_),
11     geschlecht = factor(geschlecht),
12     bildung = factor(bildung),
13     haushaltseinkommen = factor(haushaltseinkommen),
14     region = factor(region)
15 ) %>%
16 filter(complete.cases(
17     pop, OpennessMittelwert, ConscientiousnessMittelwert,
18     ExtraversionMittelwert,
19     AgreeablenessMittelwert, NeuroticismMittelwert, LiRe, alter,
20     geschlecht, bildung, haushaltseinkommen, region, ipf_gewicht))
21 df2 <- df2 %>%
22     mutate(
23         GALTAN = (galtan - 1) / 10 # 0 = GAL, 1 = TAN)
24
25 m2_galtan <- glm(
26     pop ~ OpennessMittelwert + ConscientiousnessMittelwert +
27         ExtraversionMittelwert +
28         AgreeablenessMittelwert + NeuroticismMittelwert +
29         LiRe + GALTAN + alter +
30         geschlecht + bildung + haushaltseinkommen + region,
31     data = df2,
32     family = binomial(link = "logit"),
33     weights = ipf_gewicht
34 )
35 summary(m2_galtan)
36 library(stargazer)
37
38 anova(m2, m2_galtan, test = "Chisq")
39
40 library(MASS)
41
42 # Stepwise AIC model selection starting from the full model
43 m2_selected <- stepAIC(
44     m2_galtan,
45     direction = "both", # allow adding/removing variables
46     trace = TRUE)
47
48 summary(m2_selected)
49
50 AIC(m2, m2_galtan, m2_selected)
51
52 stargazer(
53     m2, m2_galtan, m2_selected,
54     type = "latex",

```

```

55 column.labels = c("M2", "M2 + GAL-TAN", "Selected Model"),
56 dep.var.labels = "Susceptibility to populism",
57 covariate.labels = c(
58     "Openness",
59     "Conscientiousness",
60     "Extraversion",
61     "Agreeableness",
62     "Neuroticism",
63     "Left-right self-placement",
64     "GAL-TAN",
65     "Age",
66     "Gender (female)",
67     "Education: ISCED 3",
68     "Education: ISCED 4",
69     "Education: ISCED 5-8",
70     "Income: difficult to make ends meet",
71     "Income: fairly difficult to make ends meet",
72     "Income: fairly easy to make ends meet",
73     "Income: easy to make ends meet",
74     "Region (West Germany)")

```